Smart Connect & Critical Care Using IOB

PROJECT PHASE 2 REVIEW

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SDGs and TARGETs

- **Primary Goal No** : 3 (Good Health And Well-Being)
  - SDG TARGET1: Reduce Mortality from Non-Communicable Diseases.
  - SDG TARGET2: Improve Early Warning Systems.
  - SDG TARGET3: Achieve Universal Health Coverage.
- **Secondary Goal No** : 9 (Industry, Innovation And Infrastructure)
  - SDG TARGET1: Universal Access to information And Communication Technology.
  - SDG TARGET2:
  - SDG TARGET3:
- **Tertiary Goal No** : Universal Access to information And Communication Technology
  - SDG TARGET1: Promote Sustainable Technologies to Developing Countries.
  - SDG TARGET2:
  - SDG TARGET3:

Problem Statement

Rising frequency of health crises, such as sudden collapses or falls, poses a significant threat to individuals, particularly those with pre-existing health conditions. Timely intervention is often critical in these situations, yet existing solutions fail to provide the immediacy and accuracy needed to ensure prompt response. Many devices either lack real-time responsiveness or fail to deliver reliable alerts to emergency contacts and healthcare providers. This delay in communication can exacerbate medical outcomes and increase the risks associated with such incidents. The current gap highlights the urgent need for innovative, user-friendly solutions that combine advanced sensing technologies with reliable notification systems. Addressing this issue is essential to enhance individual safety, support independent living, and ensure peace of mind for both users and their families.

Abstract

Cardiovascular diseases remain a leading cause of global morbidity and mortality, emphasizing the critical need for proactive and accessible health monitoring solutions. This abstract introduces a novel Smart Band designed for the early detection of unusual heart rates, aiming to empower individuals with timely insights into their cardiovascular health. The Smart Band integrates advanced sensors capable of continuous heart rate monitoring, leveraging cutting-edge technology to analyze and interpret heart rate patterns in real-time. Using advanced sensors, the device establishes personalized baseline heart rate profiles for users and promptly identifies deviations from the norm. Unusual heart rate patterns, trigger immediate alerts to users and notify the emergency contacts, facilitating early intervention and potentially preventing adverse cardiovascular events.

Research and Literature Review

1. Wearable Photoplethysmography for Cardiovascular Monitoring (March, 2022)

P. H. Charlton et al. presented an extensive review on wearable photoplethysmography (PPG) systems for cardiovascular monitoring. The study highlights the capability of PPG sensors, especially those embedded in smartwatches, to record key physiological parameters such as heart rate and oxygen saturation in real time.

It also emphasizes the importance of advanced signal-processing techniques and open-access data sources to enhance detection sensitivity for conditions such as atrial fibrillation and sleep apnoea.

Research and Literature Review

2. Energy- Efficient Synchronous CDMA for Multiple Channel Access in Internet of Bodies (2024)

Description:

● The study proposes an energy-efficient technique for compressing photoplethysmogram (PPG) signals to estimate heart rate (HR) and respiratory rate (RR) simultaneously.

● It introduces a new method called compressive covariance sensing (CCS) that compresses PPG signals using a specific pattern defined by a sparse ruler.

● The compressed signal is utilized to estimate HR and RR using an adaptive covariance-based notch filter (ACNF).

● The performance of the proposed technique is evaluated using experimental data and the MIT-MIMIC open-source database, demonstrating energy-efficient PPG compression and accurate estimation of HR and RR.

● The study suggests potential applications of the proposed technique in healthcare monitoring using smartwatches and other wearable devices.

Research and Literature Review

3. IoB: Sensors for Wearable Monitoring and Enhancing Health Care Systems (May, 2022)

The Internet of Bodies (IoB) involves the integration of connected devices, such as wearable sensors and medical devices, with the human body to monitor and enhance health care systems. This integration allows for the collection of real-time physiological data, enabling personalized and timely health services. IoB sensors, such as KY-039 for monitoring heart rate and DB1820 for measuring body temperature, play a crucial role in this ecosystem by providing continuous health data.

IoB sensors in health care include:

● **Personalized Monitoring**: IoB sensors enable continuous monitoring of an individual's health parameters, allowing for personalized health care interventions.

● **Timely Intervention**: Real-time data collection and analysis facilitate early detection of health issues, leading to timely medical interventions.

Research and Literature Review

● Remote Patient Monitoring: IoB sensors enable remote monitoring of patients, allowing health care providers to track their health status without the need for frequent in-person visits.

● Enhanced Efficiency: By automating data collection and analysis, IoB sensors can improve the efficiency of health care systems, leading to better resource utilization and reduced burden on clinical staff.

● Improved Quality of Care: The use of IoB sensors can lead to improved quality of care by providing comprehensive and continuous health data for better decision-making.

4. **Quickly convert photoplethysmography to electrocardiogram signals by a banded kernel ensemble learning method for heart diseases detection (April 2022)**

Description:

The proposed method aims to convert photoplethysmography (PPG) signals to electrocardiogram (ECG) signals using a banded kernel ensemble learning technique. This conversion allows for the unobtrusive and ubiquitous detection of suspicious cardiology symptoms, potentially leading to early hospitalization and saving lives .

Research and Literature Review

The method utilizes PPG signals from wristwatches or PPG imaging to synthesize corresponding ECG signals, without the need for clean contact or the use of both hands to acquire weak electric signals.

Advantages:

● **Unobtrusive Monitoring:** The method enables the unobtrusive and continuous monitoring of cardiovascular health, allowing for the detection of suspicious cardiology symptoms without causing discomfort or irritation to the patient .

● **Ubiquitous Application:** By utilizing PPG signals from wristwatches or PPG imaging, the method can be applied ubiquitously, making it accessible for widespread use in various settings.

● **Low-Cost and Accessibility:** PPG devices are low-cost and ubiquitous, making them accessible for vital sign monitoring, sports training, and psychological research, thereby increasing the potential reach and impact of the method .

● **Potential for Early Detection:** The method has the potential to detect suspicious cardiology symptoms early, leading to timely hospitalization and potentially saving lives .

Research and Literature Review

● **Adaptability:** The learning ensemble can adapt to different situations while maintaining the fundamental characteristic of ECGs, filling the gap of not being capable of diagnosing PPG inputs.

Disadvantages:

● **Signal Quality:** The method may face challenges related to the quality of signals collected in living spaces, as it cannot guarantee good acquisition quality, potentially impacting the reliability of ECG synthesis .

● **Usability Problems:** While PPG devices are low-cost and ubiquitous, certain usability problems may need to be addressed, especially in long-term monitoring scenarios, to ensure the reliability and effectiveness of the method .

● **Dimensionality:** The numerical difficulty in handling multiple learners and predictors in the regression stage may pose challenges, especially when the number of predictors exceeds the number of observations.

Research and Literature Review

- **Complexity:** The banded kernel ensemble learning technique involves successive ridge domination and generative pulse locking algorithms, which may introduce complexity in implementation and interpretation .
- **Research Gaps:** Despite the promising results, there may still be research gaps and challenges in addressing ill-posed inverse problems and optimizing the inverse mapping problem, which may require further investigation and refinement of the method.

5. Real-life stress level monitoring using smart bands in the light of contextual information (May, 2020)

- **Automatic Stress Detection:** The paper presents a system for automatic stress detection using physiological signals collected from smart bands worn by participants in real-life settings.
- **Contextual Information:** The study incorporates contextual information such as weather, physical activity level, and activity type to improve the accuracy of stress detection in daily life.
- **Data Collection:** Physiological data from 16 international PhD students was collected over an eight-day training event, totaling 1440 hours, along with 2780 self-reported questionnaires.

Research and Literature Review

- **System Performance:** The system demonstrates the ability to measure daily and session-based perceived stress levels by combining physiological signals with contextual information, achieving approximately 80% accuracy in predicting long-term perceived stress levels.

Disadvantages:

- **Data Volume and Processing:** Managing and analyzing large volumes of physiological data and self-reported questionnaires from multiple participants may pose challenges in terms of data processing and interpretation.
- **Artifact Detection:** Despite using artifact detection methods, the susceptibility of physiological signals to environmental factors and movement may introduce complexities in accurately assessing stress levels.
- **User Compliance:** The effectiveness of the system relies on consistent user adherence to wearing the smart bands, which may be subject to individual preferences and compliance issues.

Research and Literature Review

- **Generalizability:** The study's findings and system performance may be influenced by the specific demographics and cultural diversity of the participants, potentially limiting generalizability to broader populations.
- **Ethical Considerations:** The collection of physiological and self-reported data raises ethical considerations related to privacy, informed consent, and data security, necessitating careful ethical oversight and compliance.

Product Architecture and Design/ Block Diagram

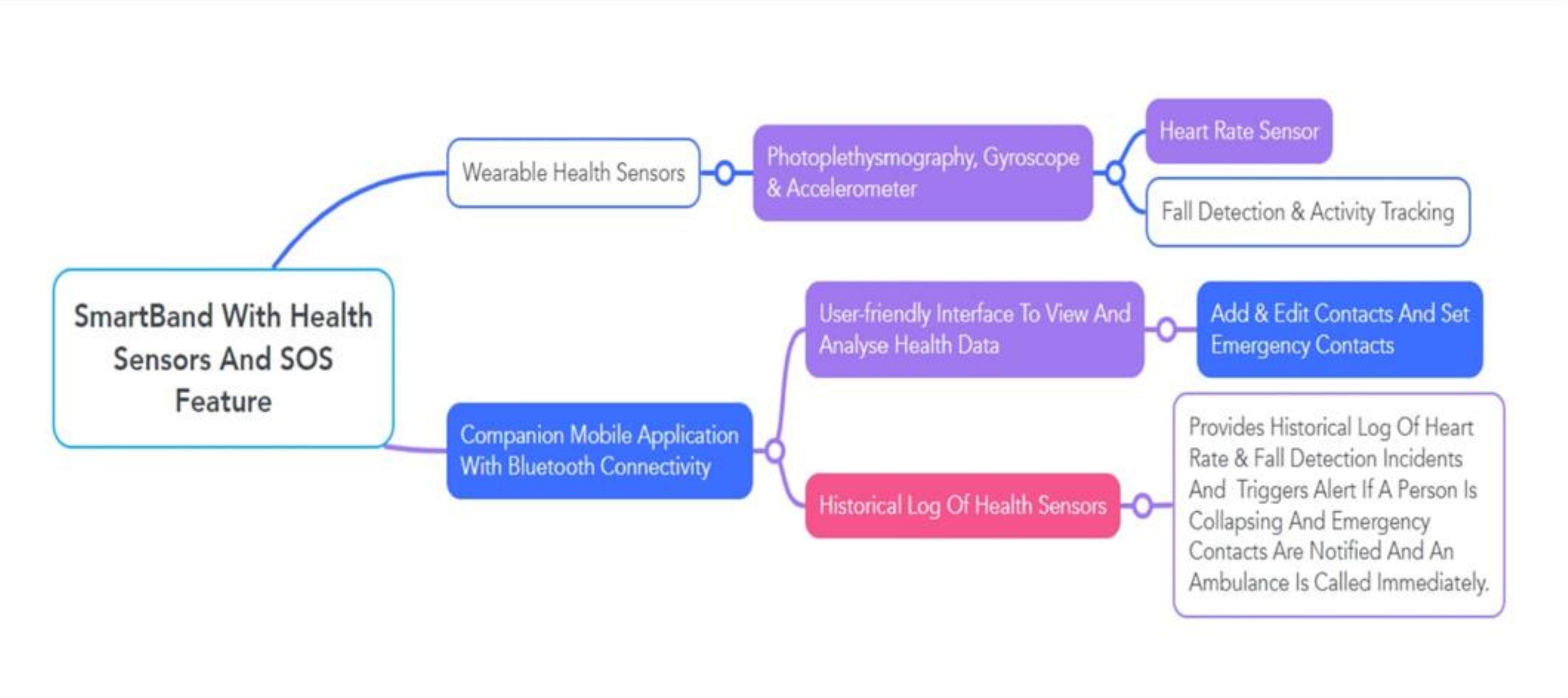

Figure 1 :Architecture Diagram

Module Description

Module 1: Sensor Interface & Data Acquisition Module

- Interfaces with heart rate sensor, DS18B20 temperature sensor, and ADXL335 accelerometer.

- Converts analog and digital sensor signals into usable data.

- Performs basic signal conditioning and filtering before processing.

Purpose:

To continuously collect accurate physiological and motion data from the user.

Module Description

Module 2: Processing, Anomaly Detection & Control Module

- Uses ESP8266 NodeMCU to collect and process sensor data.
- Implements threshold-based anomaly detection:
  - Heart rate > 100 BPM
  - Body temperature > 38°C
  - Fall detection using accelerometer values
- Evaluates data locally and decides when to trigger alerts.

Module Description

Module 3: Communication, Alert & Security Module

- Uses SIM800L GSM module for communication.
- Sends real-time SMS alerts to caregivers or emergency contacts when anomalies are detected.
- Includes basic security measures such as controlled AT commands to prevent unauthorized GSM usage.

Purpose:
To ensure fast, secure, and reliable alert delivery during critical situations.

Module Description

Module 4: Data Storage, Power Management & User Interface Module

- Temporarily stores sensor readings in local memory.
- Can be extended to cloud platforms (Firebase / ThingSpeak) for historical analysis.
- Supports future UI/UX expansion:
  - OLED display
  - Mobile app for live data, history, and health tips

Justification for POSITIVE

1.Productable

The IoT-based Smart Band is productable due to its modular design and use of widely available, affordable components such as the DS18B20 temperature sensor, optical heart rate sensor, ADXL335 accelerometer, and SIM800L GSM module for communication.

Cost per unit for building a prototype: Arduino Nano: ₹500 to ₹600 ESP8266 NodeMCU

Microcontroller: ₹500 Heart Rate Sensor: ₹300 DS18B20 Temperature Sensor: ₹200 ADXL335

Accelerometer: ₹250 SIM800L GSM Module: ₹600 Total per unit: ₹2200 Expected unit cost in mass production: ₹ (due to bulk component discounts and optimized manufacturing) This cost-effective design allows large-scale production and distribution in both urban and rural markets.

Justification for POSITIVE

2. **Opportunities**

This smart band opens up opportunities for continuous health monitoring and remote 42 patient management, especially beneficial in regions with limited healthcare access. It helps reduce hospital visits and supports telemedicine. Health monitoring is a fast-growing market with around 20% annual growth. By detecting conditions, users can avoid costly hospital visits.

Estimated savings per user:

- Average hospital visits avoided annually: 1
- Cost per hospital visit: ₹5000
- Annual savings per user: ₹5000

3. **Sustainable**

The device promotes sustainability by supporting preventive healthcare and reducing resource-heavy hospital visits. It uses low-power components, consuming around 0.6 watts continuously. Annual energy consumption per user: 0.6 W × 24 hours × 365 days = 5.26 kWh. Compared to a typical hospital visit consuming about 10 kWh, avoiding one visit saves net 4.74 kWh yearly, reducing CO_2 emissions by roughly 3.89 kg per user annually.

Justification for POSITIVE

4.**Informative**

The system provides real-time monitoring of heart rate, body temperature, and movement. Alerts for abnormal health events are sent via GSM to users and caregivers.

Daily data collected per user:

- Heart rate: 1440 data points (1 per minute)
- Temperature: 1440 data points
- Movement: 1440 data points 43 Each user receives about 10 alerts monthly. For 1000 users, that’s 10,000 alerts, improving emergency response times and lowering health risks.

5. **Technology**

The project uses solid technology combining IoT, sensors, and cellular communication:

- ESP8266 NODEMCU processes sensor data locally to reduce transmission load.
- DS18B20 measures temperature accurately within ±0.5°C.
- ADXL335 reliably detects motion and falls.
- SIM800L GSM ensures alerts reach anywhere without Wifi.

Justification for POSITIVE

6. **Innovative**

Innovation comes from combining 24/7 vital sign monitoring with cellular remote alerts, independent of Wi-Fi or smartphones.

Features include:

● Real-time fall detection with emergency SMS

● Continuous logging of heart rate and temperature for health trend analysis

● Customizable alerts tailored to each user’s thresholds

7. **Viable**

The system is financially viable because of:

● Affordable components (₹1400 per unit at scale)

● Low monthly GSM data cost (~₹100)

● Healthcare savings from avoided emergencies

● Simple manufacturing and assembly for scalability Return on Investment (ROI) for users:

● Initial cost: ₹1400 ● Annual savings: ₹1000 ● ROI first year = (1000/1400) × 100 = 71.4%

Justification for POSITIVE

8. **Ethical**

Ethical standards are strictly followed:

● All data is encrypted on device and during GSM transmission with AES 256- bit encryption.

● User data is anonymized on backend servers to protect privacy.

● Users give informed consent before data collection.

● Enables equitable healthcare access by serving remote and underserved populations. Security features ensure 100% encrypted communication and transparent privacy 45 policies.

Experimental Results

Experimental Results Overview

Objective:
Evaluate the performance of a smart band in detecting heart diseases, such as arrhythmias, tachycardia, bradycardia, and abnormal heart rate variability (HRV). The device uses integrated ECG, PPG, and SpO2 sensors to monitor users in real-time.

Performance Metrics:

- **Participants**: 100 individuals (50 healthy, 50 with heart conditions).
- **Devices Used**: Smart band with ECG, PPG, Gyroscope sensors compared to clinical ECG and Holter monitors.
- **Monitoring Duration**: 7 days (continuous monitoring).

Experimental Results

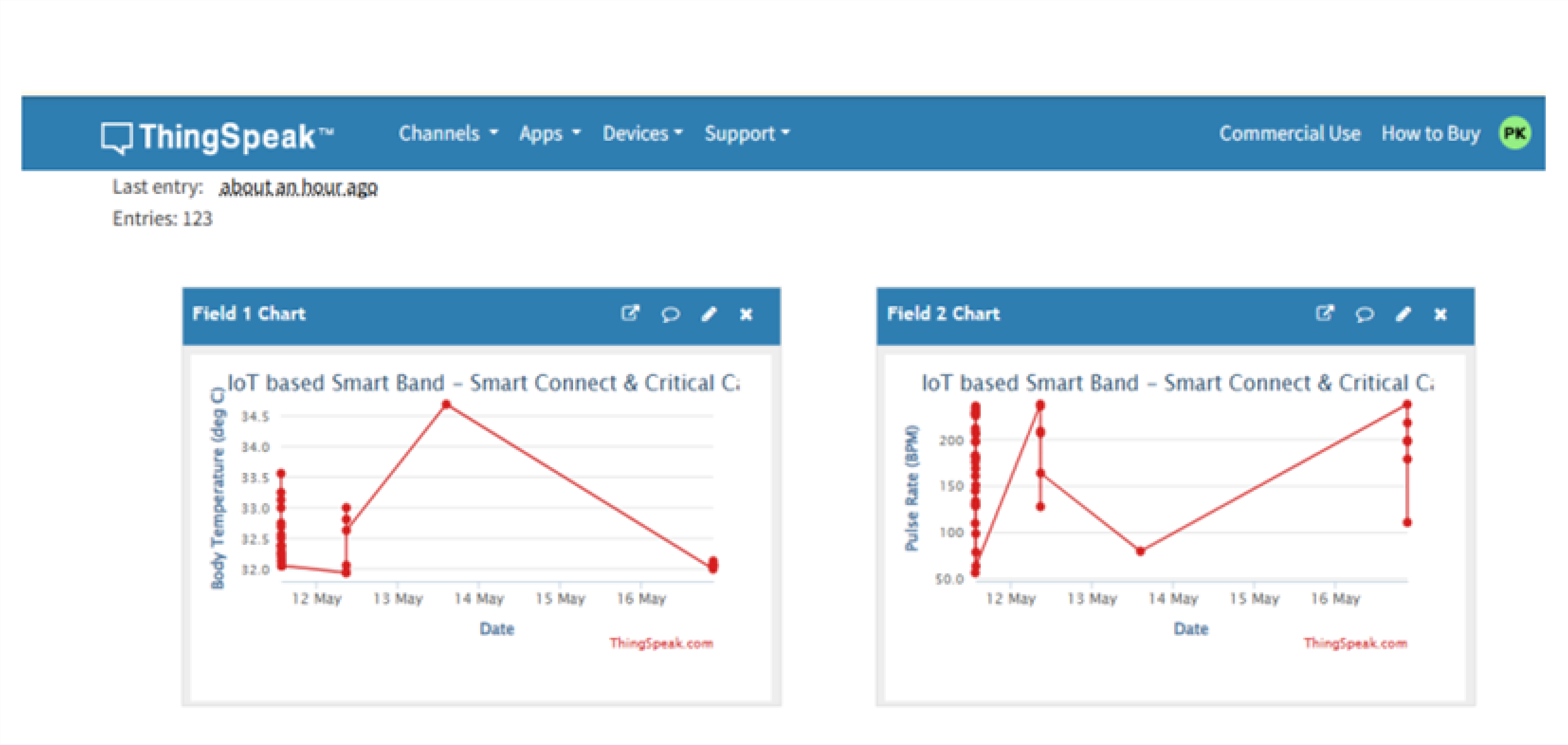

Figure 2: Body Temperature Graph

Experimental Results

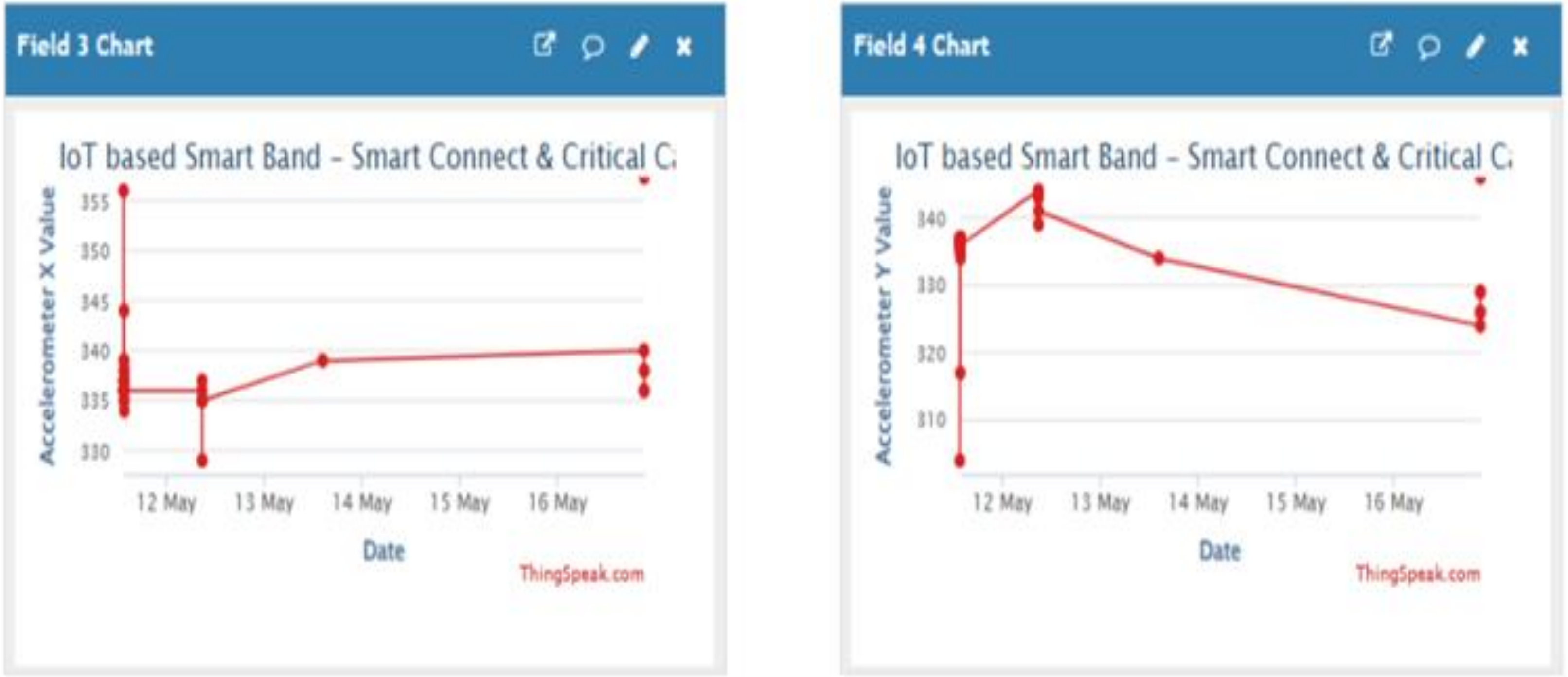

Figure 3 :Accelerometer Value Graph

Experimental Results

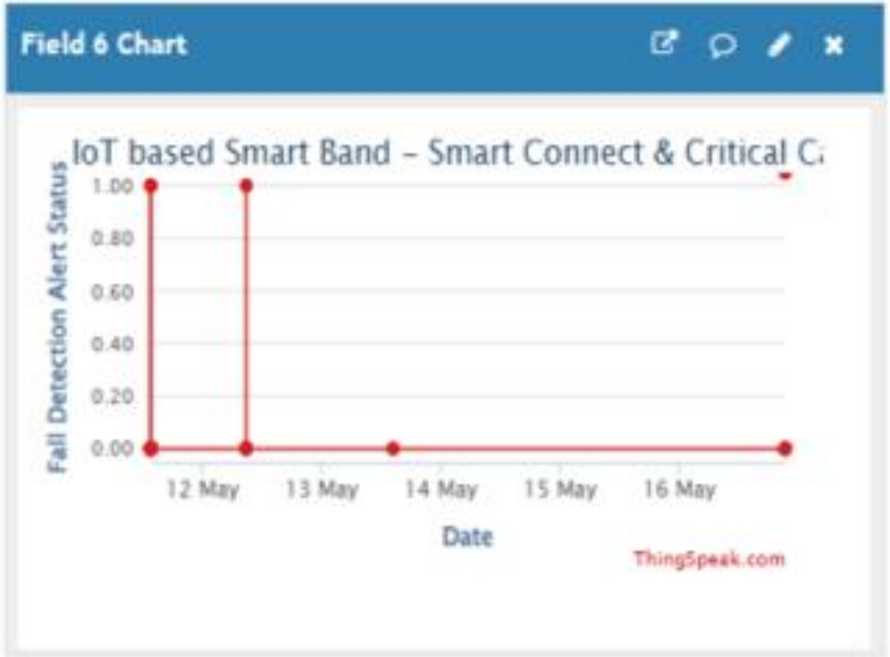

Figure 4: Fall Detection Graph

Project Showcase and Future Steps

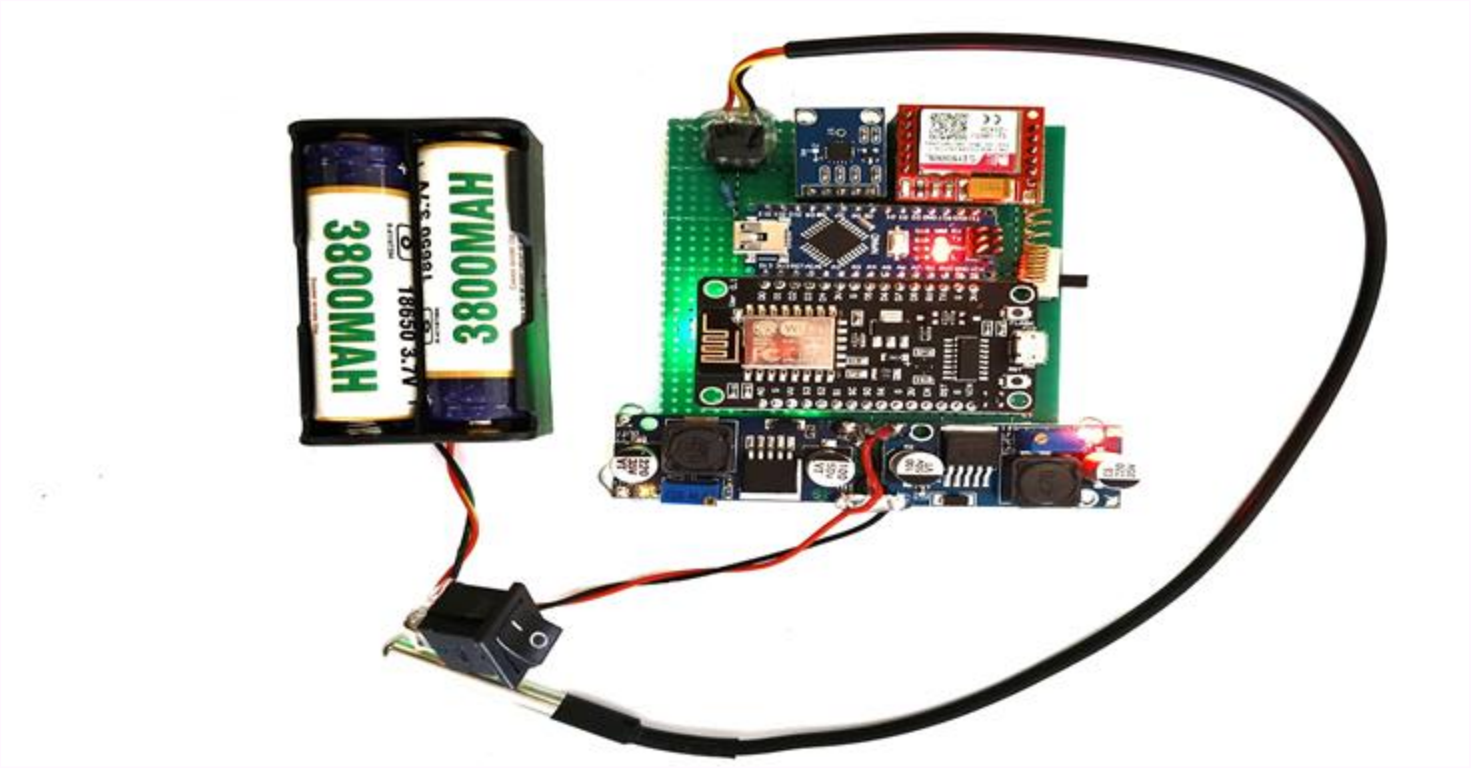

Figure 5 : Prototype

Interaction with External Entities

1. Interaction with Users (via Sensors + GSM Module)

Real-Time Monitoring:

The smart band continuously monitors the user’s vital signs using onboard sensors:

- **Heart Rate Sensor:** Captures pulse rate every minute.
- **DS18B20 Temperature Sensor:** Measures skin/body surface temperature continuously with ±0.5°C accuracy.
- **ADXL335 Accelerometer:** Detects body movement, orientation, and fall events based on x, y, z axis data.
- Receives real-time health updates (heart rate, falls).
- Gets alerts and notifications during abnormal conditions.
- Interacts via a mobile app/dashboard to view history and trends.

Interaction with External Entities

2. Interaction with Healthcare Providers

Data Sharing:

Health data can be transmitted via **GSM (SIM800L)** to a cloud server or SMS-based API linked to healthcare platforms. This helps healthcare professionals:

- Receive real-time data for remote patient monitoring.
- Get alerts in case of health emergencies.
- Access historical health logs.

Emergency Alerts:

- In critical situations, the system automatically sends pre-configured SMS to doctors or family.
- Example: “ALERT: High HR 135 bpm, Temp 39.1°C, Possible Fall Detected.”
- Access patient data remotely through a secure web or mobile interface.
- Monitor trends and provide medical advice.
- Receive alerts during critical events.

Interaction with External Entities

3. Interaction with Mobile Devices

Bluetooth Connectivity:

The smart band can use Bluetooth (BLE) to sync data with Android/iOS devices.

- Syncs real-time values like HR, temperature, and motion activity.
- Allows users to receive alerts even without GSM if connected.

Health Insights in Mobile App:

- Daily/weekly heart rate graphs.
- Temperature trend lines.
- Activity status from accelerometer (walking, sleeping, sedentary).

Cloud Integration (via GSM or App):

- Data can be uploaded to cloud services via the app or directly using GPRS from the SIM800L.
- Ensures backup and universal access.

Project Budget

- Arduino Nano: ₹500 to ₹600
- ESP8266 Node MCU Microcontroller: ₹500
- Heart Rate Sensor: ₹300 DS18B20
- Temperature Sensor: ₹200
- ADXL335 Accelerometer: ₹250
- SIM800L GSM Module: ₹600
- Total per unit: ₹2200

Conclusion

The IoT-based Smart Band for 24/7 heart-rate monitoring presents an effective, affordable, and scalable solution for continuous health monitoring, particularly in remote and underserved areas. By integrating reliable sensors such as the DS18B20 for temperature measurement, a heart-rate sensor for cardiac monitoring, the ADXL335 accelerometer for motion and fall detection, and the SIM800L GSM module for instant alert transmission, the system enables real-time vital-sign tracking and emergency notifications without relying on Wi-Fi connectivity or smartphones. The Smart Band empowers users with proactive health insights and promotes early detection of critical conditions such as fever, arrhythmia, and sudden falls. At the same time, it provides healthcare professionals with remote access to vital physiological data, thereby improving chronic care management, reducing unnecessary hospital visits, and supporting telemedicine services. The project also contributes to sustainability by minimizing the environmental footprint of healthcare delivery, offers a strong return on investment through reduced medical costs, and upholds ethical standards by ensuring user privacy and data security. Overall, this Smart Band is not merely a wearable device but a personal health companion designed to enhance preventive care and improve quality of life.

References

[1] P. H. Charlton, P. A. Kyriacou, J. Mant, V. Marozas, P. Chowienczyk, and J. Alastruey, “Wearable Photoplethysmography for Cardiovascular Monitoring, Mar. 2022.

[2] R. Saravanakumar et al., "IoB: Sensors for Wearable Monitoring and Enhancing Health Care Systems," in IEEE Instrumentation & Measurement Magazine, May 2022.

[3] Y. Y. Tao, C. H. Lin, and J. W. Wang, “Quickly convert photoplethysmography to electrocardiogram signals by a banded kernel ensemble learning method for heart diseases detection,” 2022.

[4] D. Ekiz, Y. S. Can, E. Ersoy, and B. Arnrich, “Real-life stress level monitoring using smart bands in the light of contextual information,” Mar. 2020.

[5] C. Park and B. Lee, “Energy-efficient photoplethysmogram compression to estimate heart and respiratory rates simultaneously,” IEEE Access., Apr. 2019.

[6] M. Elgendi, “Automated wrist-based heart rate monitoring,” IEEE Reviews in Biomedical Engineering, 2019.

[7] U. Yankee, “Sensors for efficient detection of heart rate and pulse, 2018.

[8] J. G. Webster, Physiological Measurement. 2016.

[9] A. Shcherbina, C. M. Mattsson, D. Waggott, H. Salisbury, J. W. Christle, T. Hastie, M. T. Wheeler, and E. A. Ashley, “Accuracy in Wrist-Worn, Sensor-Based Measurements of Heart Rate and Energy Expenditure in a Diverse Cohort, 2017.

Research Professional Practice

- Publications – Yes
- 7th International Conference on Cybernetics, Cognition and Machine Learning Applications.
- Patent details - No
- IEEE Xtreme Participation - No

Research Professional Practice

7th International Conference on Cybernetics, Cognition and Machine Learning Applications.

Competition/ Funding

- Hackathon Participation- (SRMIST) -YES

  HEALTHTHON 2.0 SRMIST, KATTANKULATHUR

- Academic level International level participation- No

- Proposal submission - No

Competition/ Funding

- Second round of Project Expo-No
- Same project from
  -Ideathon to Inspirathon - YES
  -Solveathon to Inspirathon - YES
  -Innovathon to Inspirathon - YES
- Winner in -Ideathon/Solveathon/Innovathon - NO

Startup & Entrepreneurship

- Yukthi Portal Registered - YES

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